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B																							
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D																							
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F																							
UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM SBC-W OUTGOING H04																							
				Date Prep.		15.11.2018		PROJECT TITLE				ABB		EPMV		COVER SHEET		Scale		NTS		= H04	
				Prepared		YR		MV Switchgear for MNA Serang Plant				PT. ABB Sakti Industri		MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM SBC-W OUTGOING		Customer		PT Multimas Nabati Asahan		& 305			
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant			Index of Sheet MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H04	
				Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan	& 305
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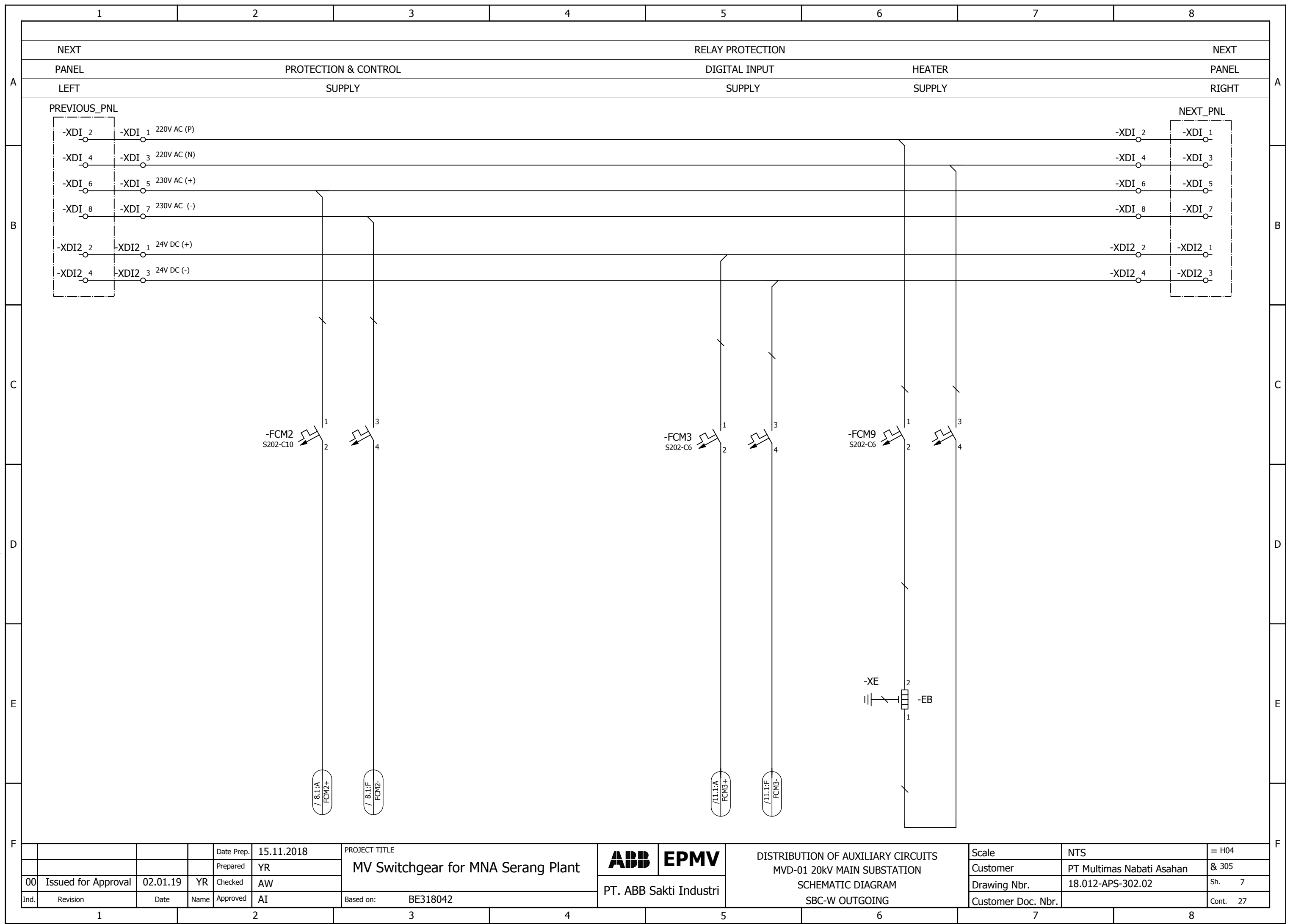
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A	-BGI31	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS							A
	-BGI32	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR							
	-BGI33	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR							
	-BGI41...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.							
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	-BGI51	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT							
	-BGI52	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT							
	-BGI53	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT							
C	-BGK	POSITION SWITCH OF KEY LOCK						-EB3	HEATER LOCATED IN MOTOR CABINET							C
	-BGL1	POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB4	HEATER FOR VENTILATION TEST							
	-BGS1...-BGS4	POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.							
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D	-BGT1	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE							D
	-BGT2	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1							
	-BGT3	POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2							
	-BGT4	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3							
E	-BGT5	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS							E
	-BGT6	POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1							
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	-BGT8	POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3							
F	-BGT11	PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER							F
	-BGT12	PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGV1	POSITION SWITCH OF FRONT VENTILATOR						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1							
	-BGV2	POSITION SWITCH OF REAR VENTILATOR						-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2							
G	-BM	HYGROSTAT						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3							G
	-BPA4	PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-FCT	THERMAL OVERLOAD RELEASES							
	-BPA5	PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS							
	-BPA6	PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR							
H	-BPA7	PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR							H
	-BPS1	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER						-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK							
	-BPS2	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2						-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE							
	-BPS3	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3						-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE							
I	-BR	FLAME DETECTORS, SMOKE DETECTORS						-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS							I
	-BT	THERMOSTAT						-KFC	CLOSING RELAYS OR CONTACTORS							
	-BT1	THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS						-KFI	INTEGRATED CIRCUITS							
	-BT2	THERMOSTAT FOR ENERGIZATION OF VENTILATORS						-KFL	LOCKOUT RELAY							
J	-BT3	THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM						-KFM	MICROPROCESSORS							J
	-BU	BUCHHOLZ RELAY						-KFN	ANTIPUMPING RELAYS							
	-BX1	UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING						-KFO	OPENING RELAYS OR CONTACTORS							
	-BX2	ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING						-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)							
K	-BZ,-BZ4	MULTIFUNCTION SENSORS						-KFR	RECLOSING RELAYS							K
	-BZ1	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1						-KFS	SYNCHRONIZING DEVICES							
	-BZ2	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2						-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS							
	-BZ3	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3						-KFZ	CONTROLLERS							
L								-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER							L
								-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD							
								-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1							
								-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2							
M								-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS							M
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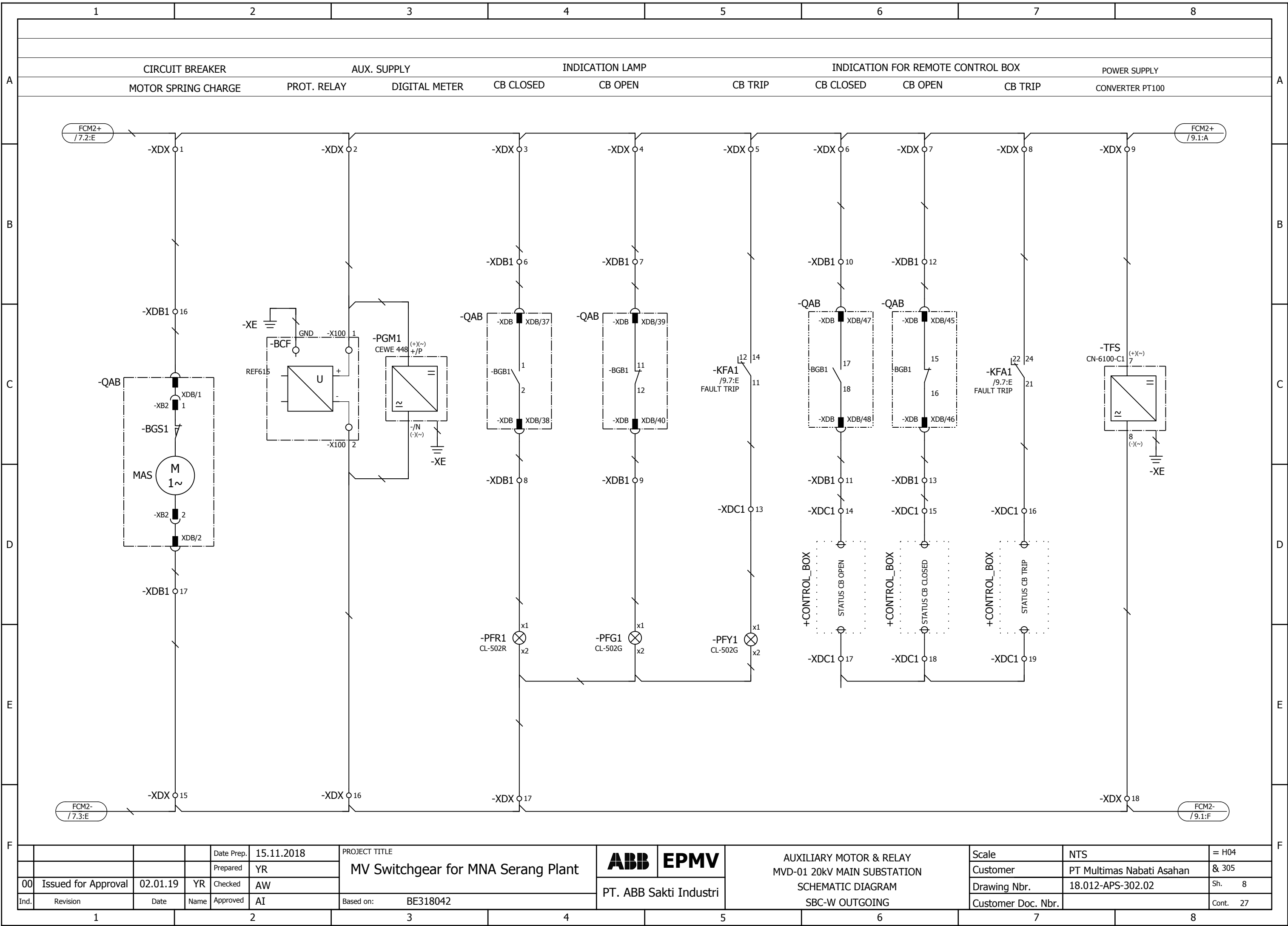
A	-MAE		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH		MANUAL CIRCUIT-BREAKERS						A
	-MAE1		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL		LINKS						
	-MAE8		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM		MINIATURE SWITCH-DISCONNECTORS						
B	-MAS		MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS		SWITCH-DISCONNECTORS						B
	-MAT		MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT		TRUCK						
	-MBC		CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA		EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING						
C	-MBC4		CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE		EARTHING SWITCH						C
	-MBO1		FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1		EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)						
	-MBO2		SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2		EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)						
D	-MBO3		OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ		CLUTCHES						D
	-MBO4		OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA		ANTIFERRORESONANCE RESISTOR						
	-MBU		UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD		DIODES						
E	-PFB		BLUE SIGNAL LAMPS						-RAI		INDUCTORS						E
	-PFF		FLAG RELAYS						-RAR		RESISTORS						
	-PFG		GREEN SIGNAL LAMPS						-RF		FILTERS						
F	-PFR		RED SIGNAL LAMPS						-RLE1		ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING						F
	-PFS		SHORT CIRCUIT INDICATORS						-RLE2		ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT						
	-PFV		VOLTAGE INDICATORS						-RLE3		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE						
G	-PFV1		VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4		ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION						G
	-PFV2		VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5		ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD						
	-PFW		WHITE SIGNAL LAMPS						-RLE6		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1						
H	-PFX		CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2						H
	-PFY		YELLOW SIGNAL LAMPS						-RLE8		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						
	-PGA		AMMETERS						-RLE9		ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS						
I	-PGC		COUNTERS						-RLE11		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4						I
	-PGF		FREQUENCYMETERS						-RLE12		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5						
	-PGH		HOURMETERS						-SFA		AMMETRIC SWITCHES						
J	-PGI		PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC		CONTROL SWITCHES, CLOSING PUSH-BUTTONS						J
	-PGI1		HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL		LOCKING CONTACTS						
	-PGI2		HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO		OPENING PUSH-BUTTONS						
K	-PGJ		ACTIVE ENERGY METERS						-SFR		RESET PUSH-BUTTONS						K
	-PGK		REACTIVE ENERGY METERS						-SFS		SELECTOR SWITCHES						
	-PGM		MULTIFUNCTION INDICATORS						-SFT		TEST PUSH-BUTTONS						
L	-PGP		POWER-FACTOR METERS						-SFU		UNLOCKING PUSH-BUTTONS						L
	-PGQ		VARMETERS						-SFV		VOLTMETRIC SWITCHES						
	-PGS		SYNCHRONOSCOPES						-TA		POWER TRANSFORMERS						
M	-PGV		VOLTMETERS						-TB		CONVERTER						M
	-PGW		WATTMETERS						-TB1		RECTIFIER FOR FIRST OPENING RELEASE						
	-PJ		ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB2		RECTIFIER FOR SECOND OPENING RELEASE						
N	-QAB		CIRCUIT-BREAKERS						-TB3		RECTIFIER FOR CLOSING RELEASE						N
	-QAC		CONTACTORS (FOR POWER)						-TB4		RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING						
	-QBD		DISCONNECTORS						-TB5		RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT						
O	-QBD1		FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6		RECTIFIER FOR UNDERVOLTAGE RELEASE						O
	-QBD2		REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7		ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						
	-QBD3		FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)														
P	-QBD4		TWO-WAY DISCONNECTOR														P
	-QBD5		TWO-WAY DISCONNECTOR ON BUSBAR RISER														
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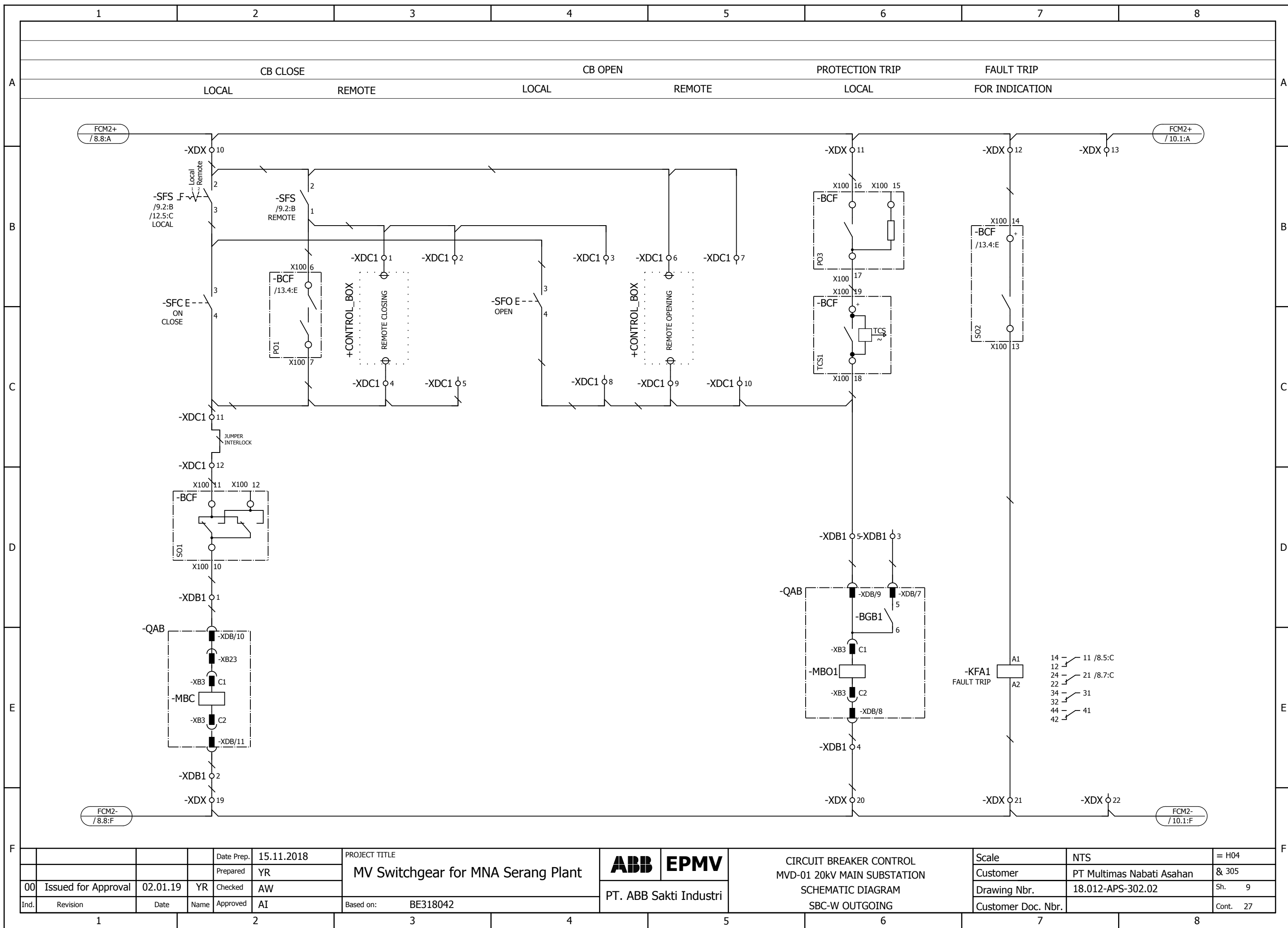
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A	-TFA	ACTIVE POWER TRANSDUCERS						-XDI1	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS							A
	-TFC	CURRENT TRANSDUCERS						-XDI2	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING							
	-TFF	FREQUENCY TRANSDUCERS						-XDI3	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION							
	-TFJ	ACTIVE ENERGY TRANSDUCERS						-XDI4	TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS							
	-TFK	REACTIVE ENERGY TRANSDUCERS						-XDI6	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS							
B	-TFM	MULTIFUNCTION TRANSDUCERS						-XDI7	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS							B
	-TFP	POWER-FACTOR TRANSDUCERS						-XDI8	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS							
	-TFQ	REACTIVE POWER TRANSDUCERS						-XDI9	TERMINAL BLOCK FOR SPECIAL USE							
	-TFS	SIGNAL CONVERTERS						-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD							
	-TFV	VOLTAGE TRANSDUCERS						-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS							
C	-WA	BUSBARS						-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT							C
	-WBC	POWER CABLES						-XDS	CURRENT SOCKETS							
	-WE	EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.						-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB							
	-WGC	CONTROL CABLES						-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK							
	-WGS	COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)						-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK							
D	-WH	OPTICAL FIBRES						-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS							D
	-XDA	TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS						-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1							
	-XDA1	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1						-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2							
	-XDA2	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2						-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3							
	-XDA3	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3						-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR							
E	-XDA5	CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER						-XDX	SUPPORT TERMINAL BLOCK							E
	-XDA8	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS						-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE							
	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)						-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE							
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER						-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE							
	-XDB10...-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS						-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE							
F	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS						-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE							F
	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR						-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE							
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR						-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT							
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR														
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET														
G	-XDC	CUSTOMER TERMINAL BLOCK														G
	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER														
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	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT														
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I	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS														I
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	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS														
	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS														
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)														
J	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)														J
					Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	REFERENCE DESIGNATIONS MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM SBC-W OUTGOING		Scale	NTS	= H04	
					Prepared	YR							Customer	PT Multimas Nabati Asahan	& 305	
	00	Issued for Approval	02.01.19	YR	Checked	AW							Drawing Nbr.	18.012-APS-302.02	Sh. 6	
	Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	PT. ABB Sakti Industri				Customer Doc. Nbr.		Cont. 27	



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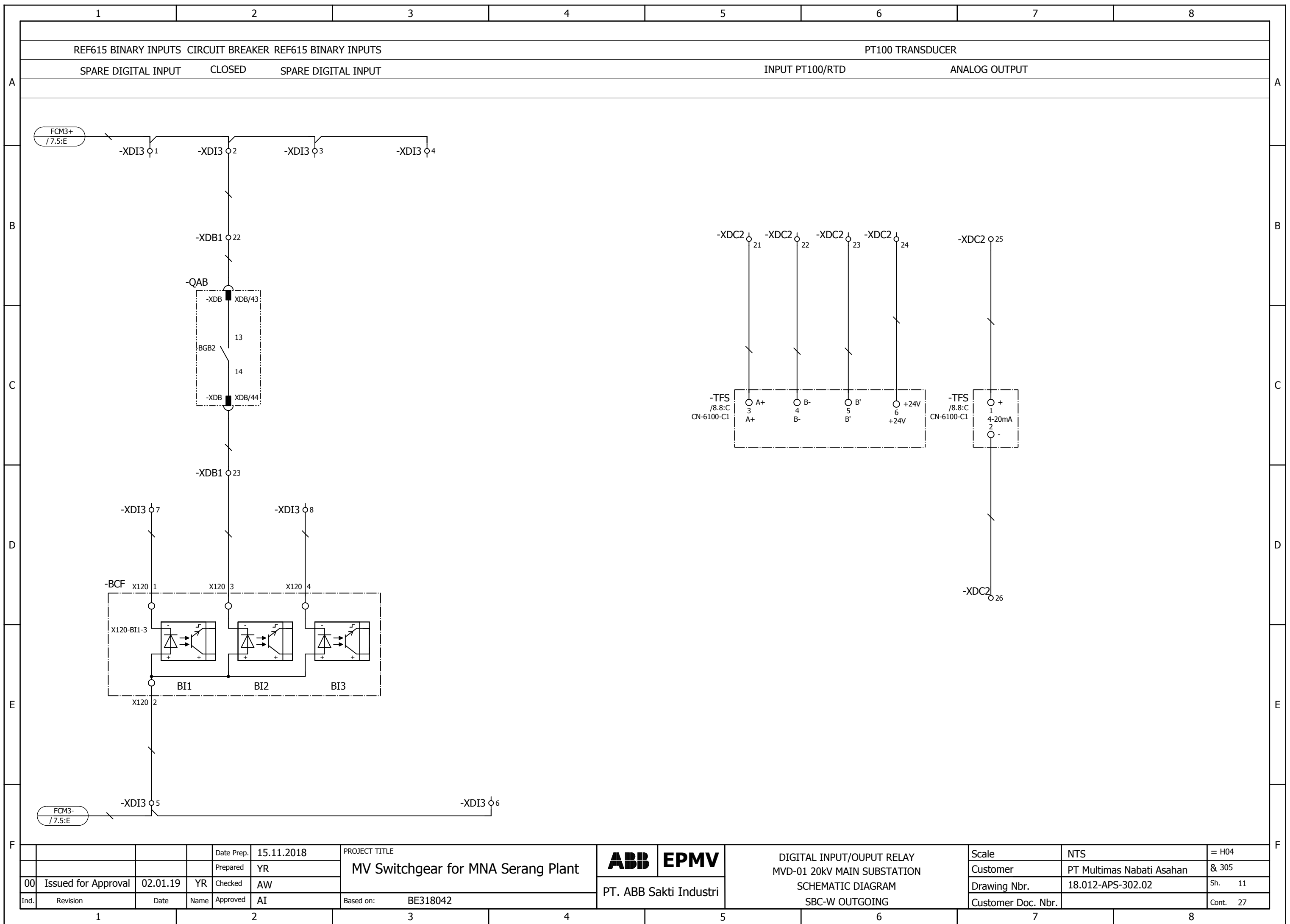


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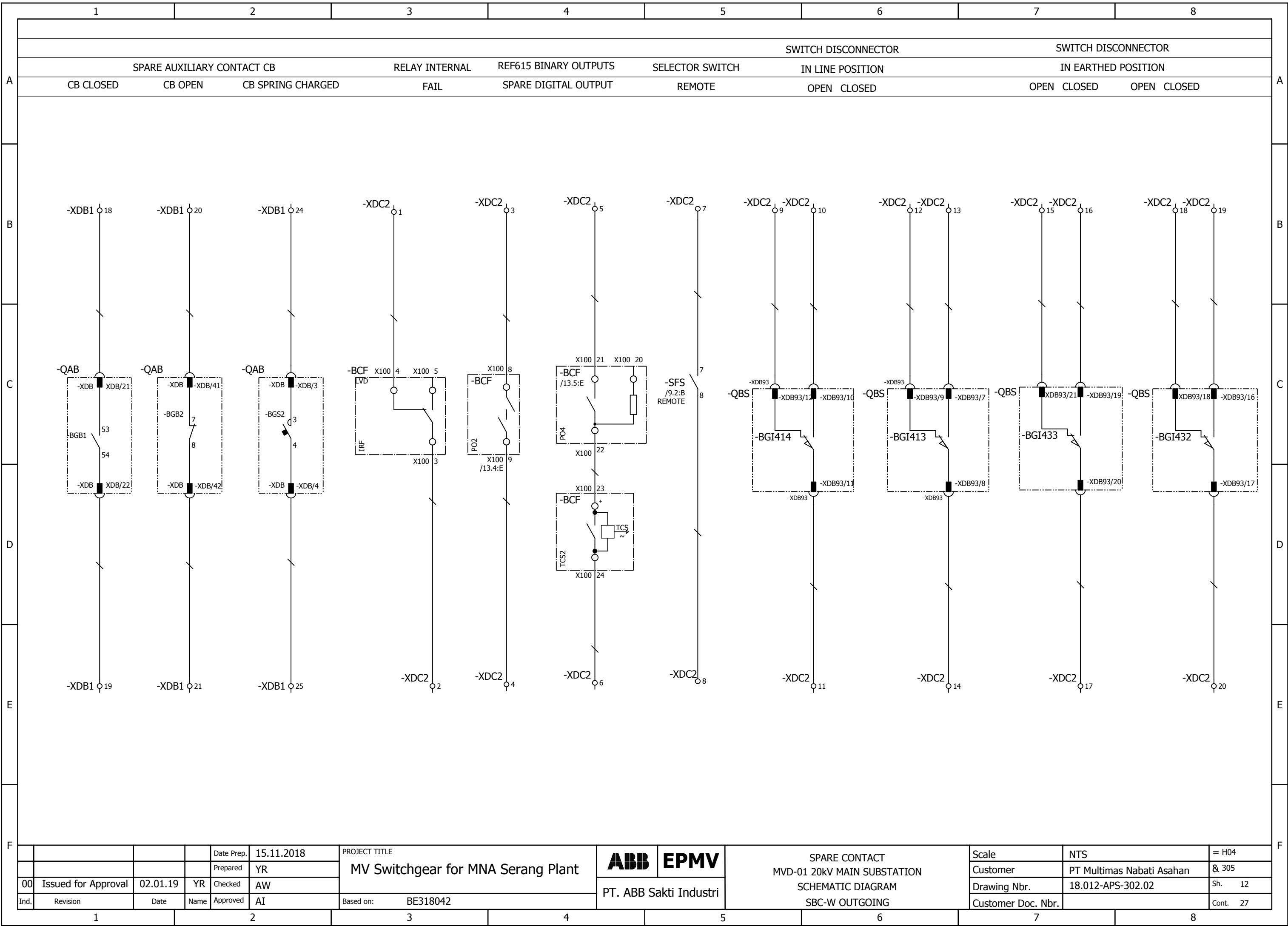


				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	AUXILIARY MOTOR & RELAY MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H04
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 305
00	Issued for Approval	02.01.19	YR	Checked	AW		Drawing Nbr.	18.012-APS-302.02		Sh. 10		
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	Customer Doc. Nbr.			Cont. 27	
						PT. ABB Sakti Industri						

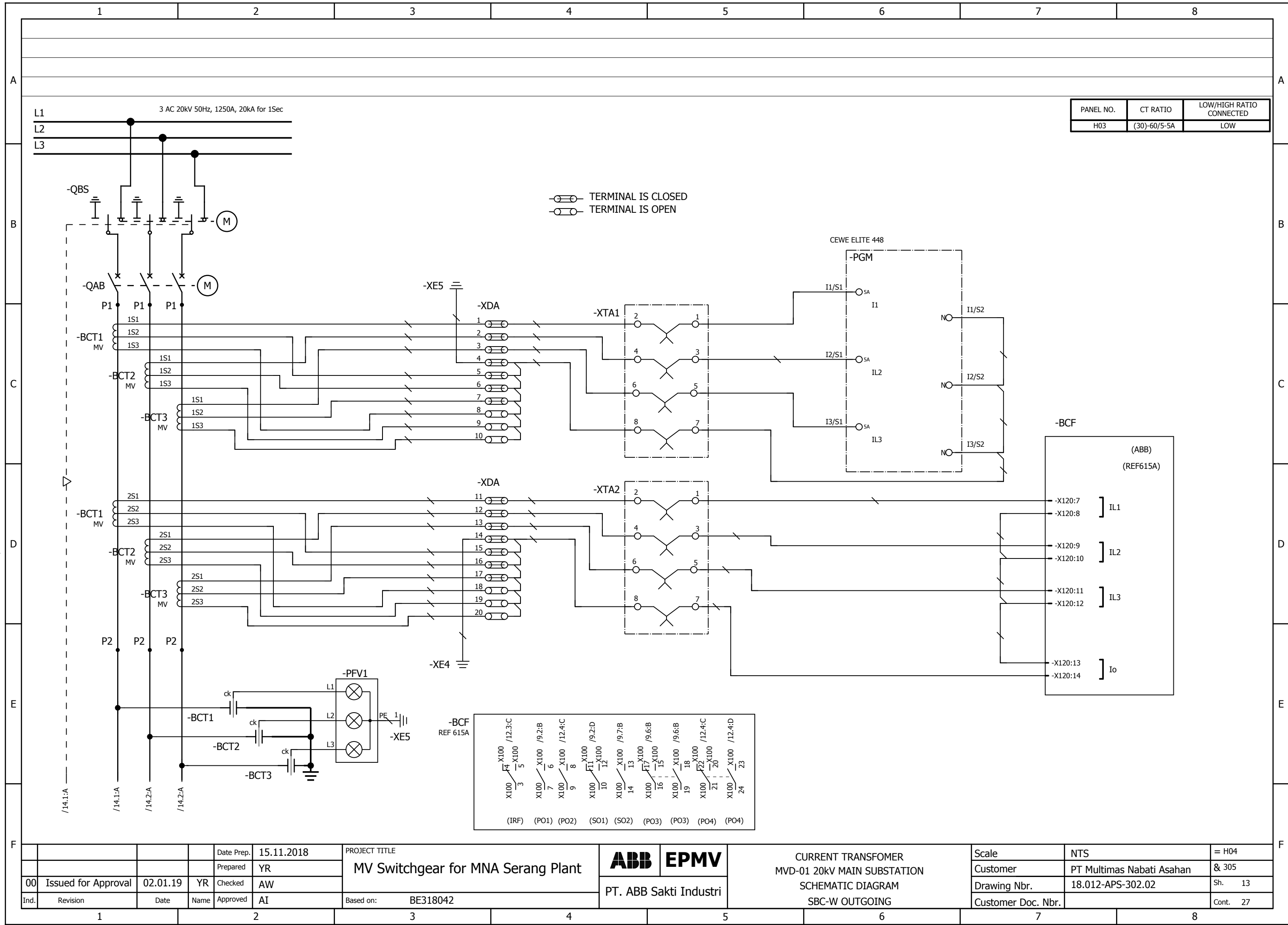
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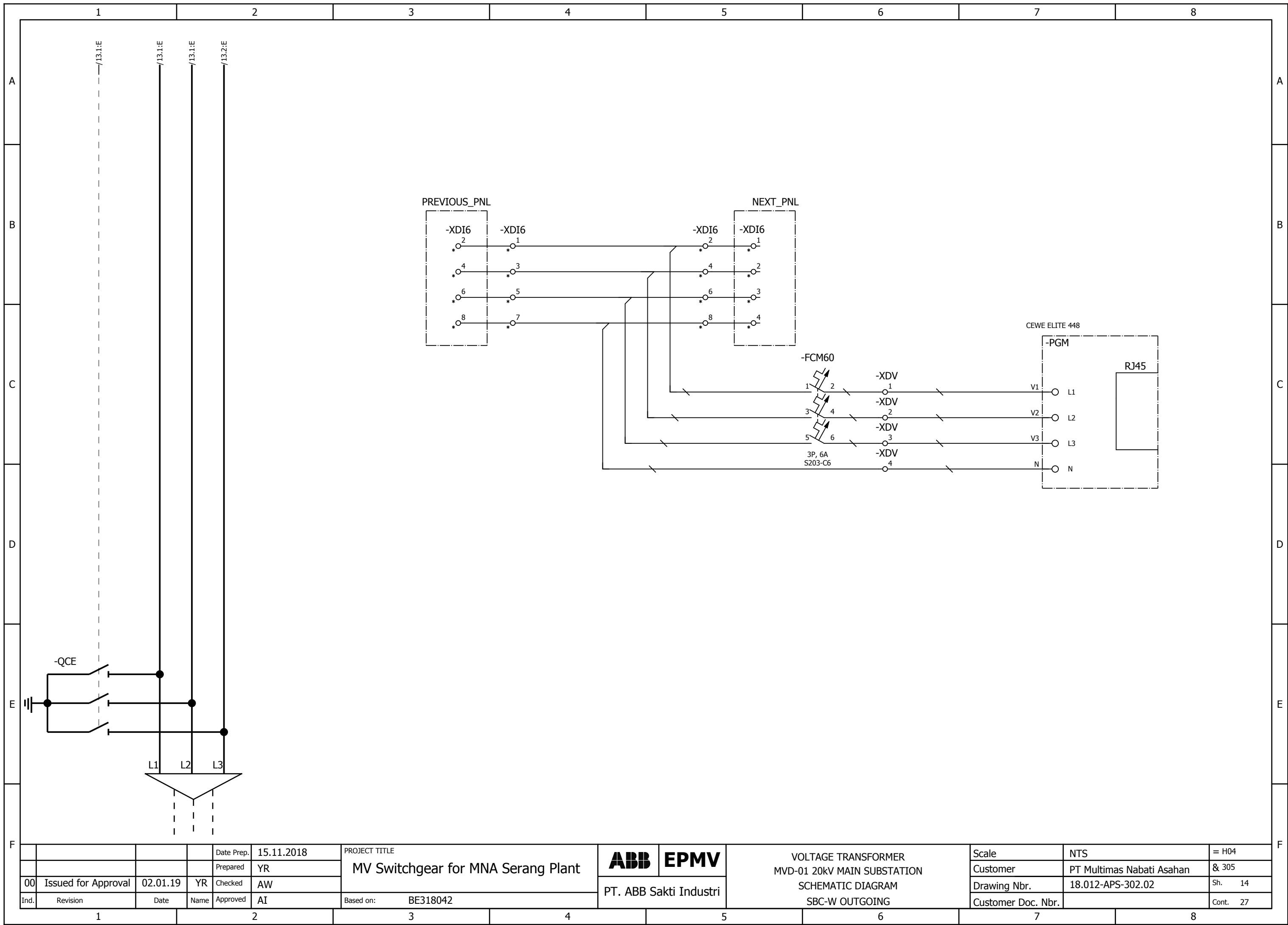


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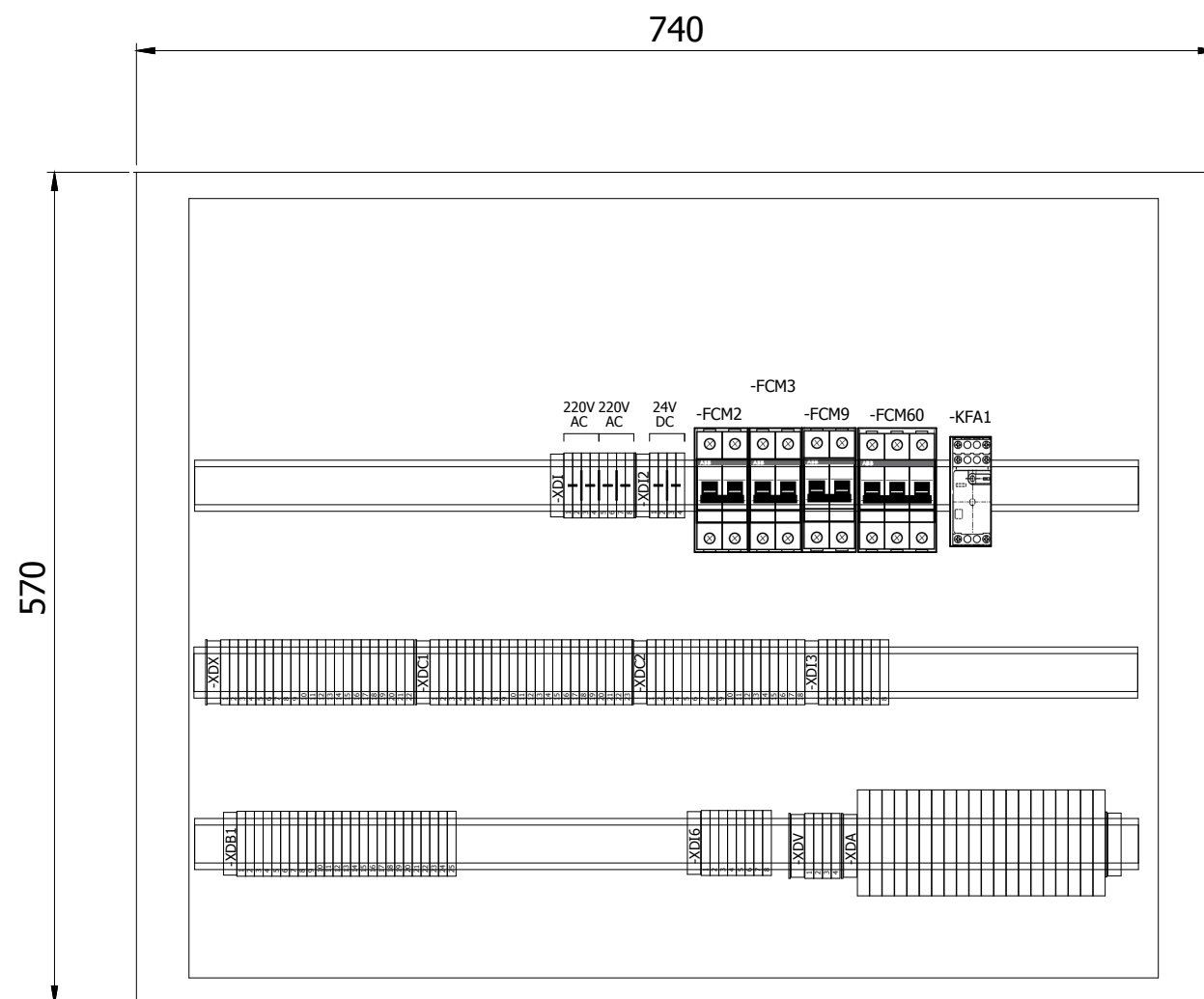


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No.	EQUIPMENT DESCRIPTION	ENGRAVING
1	-SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)	-SFS, LOCAL-OFF-REMOTE
2	-SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)	-SFC, CB CLOSE
3	-SFO (PUSH BUTTON CB OPEN, RED, ABB)	-SFO, CB OPEN
4	-PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB)	-PFY1, CB TRIP
5	-PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)	-PFR1, CB CLOSE
6	-PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)	-PFG1, CB OPEN
7	-PGJ (DIGITAL MEASUREMENT, CEWE 448)	-PGJ, DIGITAL MEASUREMENT
8	-BCF (FEEDER PROTECTION RELAY, REF615J, ABB)	-BCF, RELAY PROTECTION
9	-XTA1, XTA2 (TEST BLOCK , YONGSUNG)	-XTA, CURRENT TEST BLOCK

[illegible]

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	LV BOX LAYOUT MVD-01 20kV MAIN SUBSTATION SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H04
				Prepared	YR							Customer
00	Issued for Approval	02.01.19	YR	Checked	AW					Drawing Nbr.	18.012-APS-302.02	Sh. 16
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 27

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

F		E				D		C		B		A									
		REMARK		CABLE		EQUIPMENT		Wire No		Terminal number		TYPE		WIRE NO.		EQUIPMENT		CABLE		REMARK	
		BLACK 1,5 mm²				-PGM:V1				1		PT 2.5				-FCM60:2				BLACK 1,5 mm²	
		BLACK 1,5 mm²				-PGM:V2				2		PT 2.5				-FCM60:4				BLACK 1,5 mm²	
		BLACK 1,5 mm²				-PGM:V3				3		PT 2.5				-FCM60:6				BLACK 1,5 mm²	
		BLACK 1,5 mm²				-PGM:N				4		PT 2.5				-XDI6:8				BLACK 1,5 mm²	

F										E										D										C										B										A									
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